

EVELETH-VIRGINIA, MN, USA

WMO: 727474

Lat: 47.424N

Lon: 92.498W

Elev: 420

StdP: 96.38

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.2	-27.3	-34.1	0.2	-30.0	-31.2	0.2	-26.4	9.2	-10.2	8.3	-10.7	1.1	310	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	29.7	20.6	27.7	19.3	26.6	18.5	22.6	27.4	21.3	25.8	20.2	24.7	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.7	25.4	19.9	15.3	24.2	18.2	13.8	22.5	68.7	27.2	63.9	25.6	59.9	24.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.8	6.9	-34.7	32.3	4.0	1.9	-37.5	33.7	-39.9	34.8	-42.1	35.9	-45.0	37.2
			-34.7	24.5	3.6	1.3	-37.2	25.4	-39.3	26.2	-41.3	26.9	-43.9	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-12.8	-10.4	-3.4	4.0	10.8	16.1	18.9	17.5	13.2	6.2	-1.9	-9.3
	DBStd	12.29	7.87	7.29	7.06	5.07	4.44	3.78	3.19	3.20	4.43	4.53	5.90	7.01
	HDD10.0	3042	707	572	417	189	42	2	0	0	18	137	358	599
	HDD18.3	5292	966	806	674	432	237	86	32	53	163	378	608	858
	CDD10.0	902	0	0	1	8	67	186	274	234	113	18	1	0
	CDD18.3	111	0	0	0	0	4	19	48	29	9	1	0	0
	CDH23.3	1399	0	0	0	5	91	268	576	348	104	8	0	0
	CDH26.7	294	0	0	0	0	16	58	138	66	16	1	0	0
Wind	WSAvg	2.8	2.7	2.8	2.9	3.0	3.0	2.5	2.4	2.4	2.6	3.0	3.0	2.7
Precipitation	PrecAvg	688	22	15	29	49	73	109	103	90	83	65	34	21
	PrecMax	875	83	49	76	138	150	214	302	220	171	178	93	51
	PrecMin	409	4	0	3	7	11	30	27	15	18	8	3	2
	PrecStd	95	17	11	16	31	33	44	60	51	37	42	24	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	7.6	17.3	22.8	28.8	31.3	32.4	31.1	28.6	24.1	15.8	5.3
		MCWB	2.2	4.1	10.8	12.6	17.8	22.3	22.3	21.8	19.5	15.4	10.3	2.1
	2%	DB	1.9	4.8	12.4	18.9	26.2	27.9	29.9	27.9	26.3	19.9	12.2	2.9
		MCWB	0.7	2.3	7.4	9.9	15.6	19.2	21.2	19.6	18.5	13.4	8.7	1.4
	5%	DB	0.1	2.4	8.7	16.9	22.9	26.3	27.8	27.1	23.7	17.1	8.0	1.4
		MCWB	-0.5	0.7	5.1	8.4	13.8	17.8	19.7	19.2	17.3	12.2	5.3	0.5
	10%	DB	-2.1	0.2	6.1	14.0	20.2	24.7	26.7	25.9	22.2	14.7	6.2	0.2
		MCWB	-2.9	-0.9	2.8	6.9	12.6	17.1	19.0	18.4	16.6	10.3	3.7	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	4.5	13.0	14.2	19.8	23.7	24.3	23.7	21.9	17.2	12.1	3.5
		MCDB	4.7	6.8	15.9	20.0	25.9	28.7	29.2	28.2	26.2	21.2	13.7	4.5
	2%	WB	0.6	2.5	8.0	11.0	17.4	21.5	22.9	21.9	20.2	14.9	8.4	1.9
		MCDB	1.6	4.4	11.5	16.5	22.9	26.4	27.8	26.2	24.3	18.6	12.1	3.0
	5%	WB	-0.6	0.6	5.4	9.1	15.3	19.7	21.6	20.7	18.8	12.8	6.0	0.5
		MCDB	-0.1	2.1	8.8	15.7	21.0	24.0	26.0	25.1	22.6	16.4	8.4	1.3
	10%	WB	-2.5	-0.8	3.1	7.3	13.7	18.3	20.3	19.5	17.2	10.7	3.9	-0.3
		MCDB	-2.1	0.4	5.6	13.2	18.6	22.5	24.8	24.0	20.5	13.5	6.0	0.1
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.8	11.6	12.2	13.5	12.9	13.2	13.0	12.3	10.0	8.3	8.8
		MCDBR	9.5	12.1	13.9	18.4	18.3	15.7	15.0	15.4	15.7	15.9	12.4	9.9
		MCWBR	8.6	9.7	9.3	10.5	9.6	8.4	7.8	8.4	9.3	10.0	8.6	8.6
	5% WB	MCDBR	8.7	10.3	12.9	16.5	15.1	13.0	13.4	12.9	12.3	13.6	11.9	9.0
		MCWBR	8.1	8.5	9.1	10.1	8.8	7.9	7.9	7.9	8.0	9.5	8.6	8.0
Clear-Sky Solar Irradiance	taub	0.245	0.259	0.308	0.345	0.363	0.370	0.387	0.375	0.352	0.308	0.283	0.259	
	taud	2.357	2.325	2.344	2.378	2.386	2.387	2.358	2.421	2.460	2.572	2.520	2.396	
	Ebn at Noon	859	919	915	909	901	892	871	868	858	857	811	794	
	Edh at Noon	69	89	104	112	115	116	118	106	93	71	60	60	
All-Sky Solar Radiation	RadAvg	1.43	2.56	3.71	4.64	5.25	5.61	6.03	5.10	3.79	2.26	1.31	1.01	
	RadStd	0.17	0.24	0.35	0.47	0.38	0.42	0.27	0.39	0.29	0.31	0.17	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page